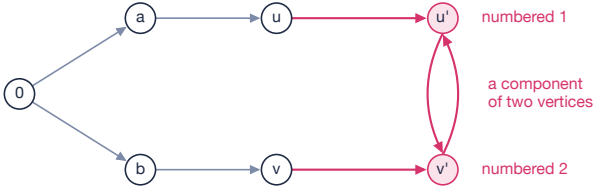
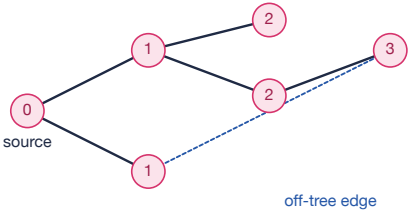
a The gadget behind Proposition 4



Whichever of u and v depth-first search reaches first steps into the new pair first, so the pair's representative (its first-discovered vertex) is u' exactly when u comes before v.

b A Kirchhoff read-out on a tree



Conductance 1 on tree edges: each potential is the node's depth, so it orients parent and child. An off-tree edge lowers the deep potentials.